

Autodesk® Scaleform®

Scale9Grid

Scaleform Scale9Grid

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: 1. 01
: 2008 3 21

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Autodesk® Scaleform® 4.3

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1. 소개: 플래시와 Scaleform 에서의 Scale9Grid

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Flash®
Scale9Grid

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MovieClip.scale9Grid Button.scale9Grid
ActionScript
Scale9Grid

Scale9Grid

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Scaleform Scale9Grid

Scaleform Scale9Grid

Scale9Grid 특성	어도비 플래시 플레이어	Scaleform
	X/Y	
		Scale9Grid
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Scale9Grids		

Scaleform
Scaleform

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Scale9Grid
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FLA/SWF

2. 변환

Scaleform Scale9Grid

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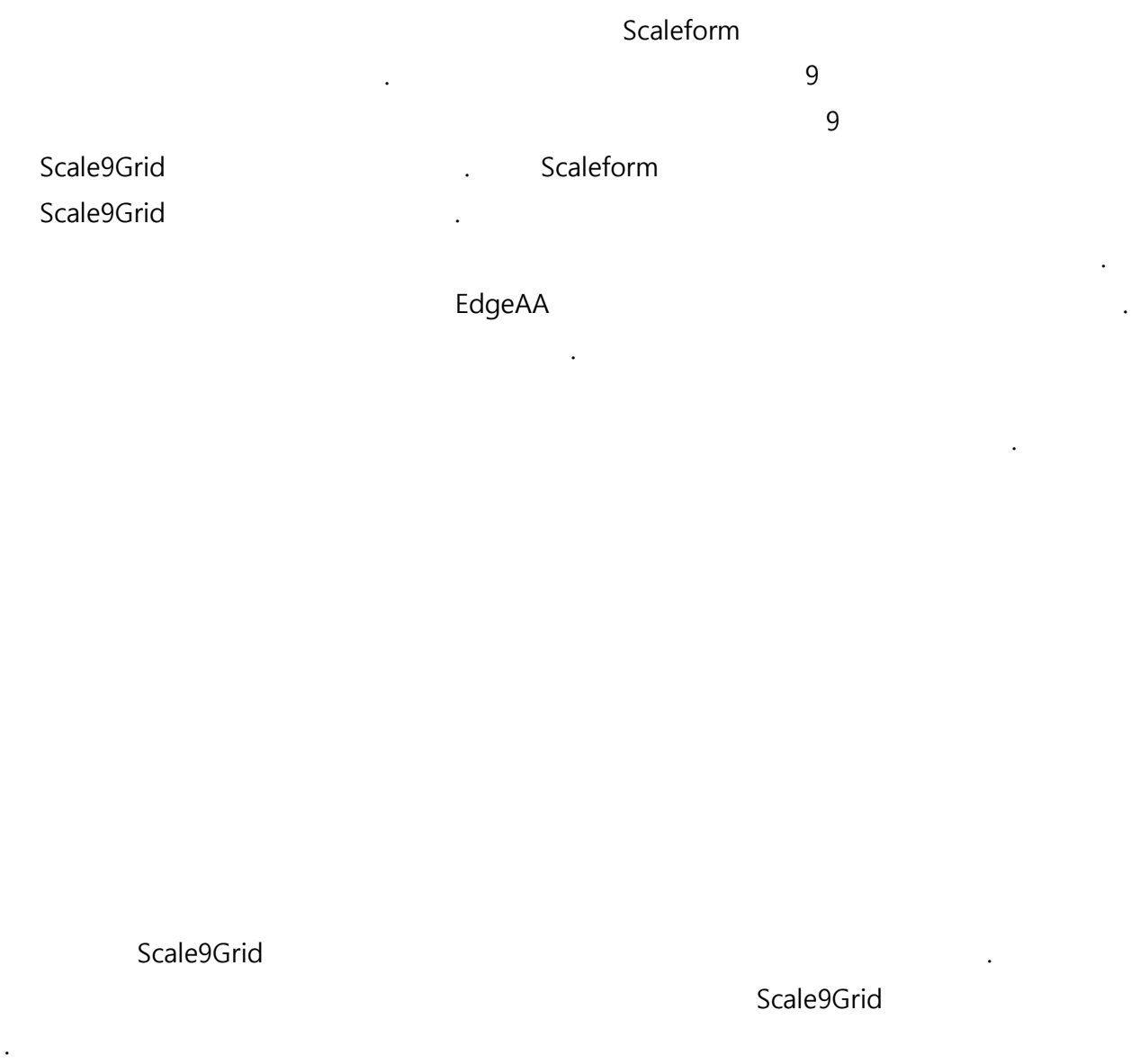
Scaleform

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Scale9Grid

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3. 비트맵 처리

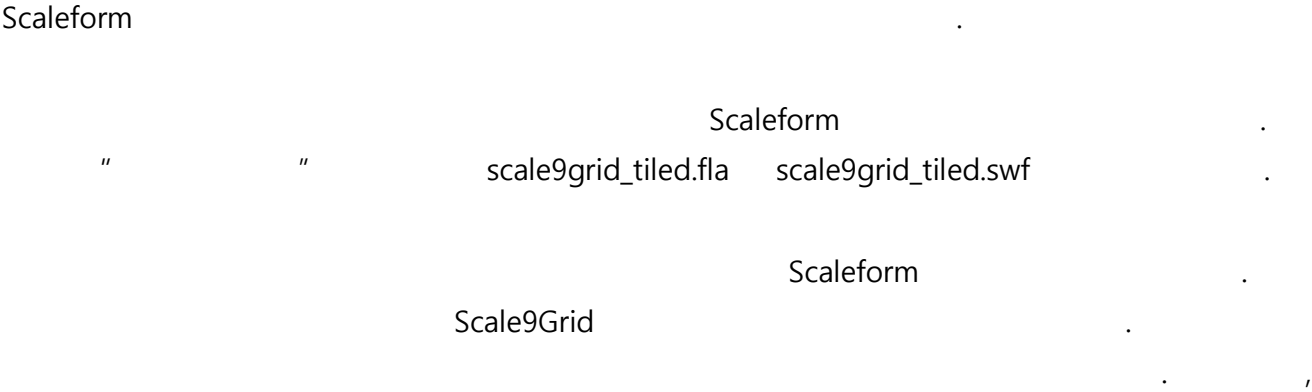


Scale9Grid

Scaleform

Scale9Grid

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9



4. 인터랙티브 윈도우

ActionScript

4.1

scale9grid_window1 fla

scale9gT906.□

P5□□

```

this.XMax = bounds.xMax;
this.YMax = bounds.yMax;
this.MinW = 120;
this.MinH = 100;
this.OldX = this._x;
this.OldY = this._y;
this.OldW = this._width;
this.OldH = this._height;
this.OldMouseX = 0;
this.OldMouseY = 0;
this.XMode = 0;
this.YMode = 0;

```

```

this.onPress = function()
{
    this.OldX = this._x;
    this.OldY = this._y;
    this.OldW = this._width;
    this.OldH = this._height;
    this.OldMouseX = _root._xmouse;
    this.OldMouseY = _root._ymouse;
    this.XMode = 0;
    this.YMode = 0;

    var kx = (this.XMax - this.XMin) / this._width;
    var ky = (this.YMax - this.YMin) / this._height;
    var x1 = this.XMin + 6 * kx; //â@w
    var y1 = this.YMin + 4 * ky; //3 %>
    var x2 = this.XMax - 26 * kx; //9 @w
    var y2 = this.YMax - 25 * ky; //G®%>
    var w = 20 * kx;
    var h = 20 * ky;
    var xms = this._xmouse;
    var yms = this._ymouse;

    if (xms >= x1 && xms <= x1+w)
    {
        if (yms >= y1 && yms <= y1+h)
        {
            this.XMode = - 1;
            this.YMode = - 1;
        }
        else
        if (yms >= y2 && yms <= y2+h)
        {
            this.XMode = - 1;
            this.YMode = 1;
        }
    }
    else
    if (xms >= x2 && xms <= x2+w)
    {
        if (yms >= y1 && yms <= y1+h)
        {
            this.XMode = 1;

```

```

        this.YMode = - 1;
    }
    else
    if (yms >= y2 && yms <= y2+h)
    {
        this.XMode = 1;
        this.YMode = 1;
    }
}

if (XMode == 0 && YMode == 0)
{
    this.startDrag();
}
}

this.onRelease = function()
{
    this.XMode = 0;
    this.YMode = 0;
    this.stopDrag();
}

this.onReleaseOutside = function()
{
    this.XMode = 0;
    this.YMode = 0;
    this.stopDrag();
}

this.onMouseMove = function()
{
    var dx = _root._xmouse - OldMouseX;
    var dy = _root._ymouse - OldMouseY;

    if (this.XMode == - 1)
    {
        this._x = this.OldX + dx;
        this._width = this.OldW - dx;
        if (this._width < this.MinW || _root._xmouse > this.OldX + this.OldW)
        {
            this._x = this.OldX + this.OldW - this.MinW;
            this._width = this.MinW;
        }
    }
    if (this.XMode == 1)
    {
        this._width = this.OldW + dx;
        if (this._width < this.MinW || _root._xmouse < this.OldX)
            this._width = this.MinW;
    }
    if (this.YMode == - 1)
    {
        this._y = this.OldY + dy;
        this._height = this.OldH - dy;
        if (this._height < this.MinH || _root._ymouse > this.OldY + this.OldH)

```

```

        {
            this._y = this.OldY + this.OldH - this.MinH;
            this._height = this.MinH;
        }
    }
    if (this.YMode == 1)
    {
        this._height = this.OldH + dy;
        if (this._height < this.MinH || _root._ymouse < this.OldY)
            this._height = this.MinH;
    }
}

```

```

        XMode YMode . -1
        . 1

```

```

        onPress()
        Scale9Grid

```

```

var kx = (this.XMax - this.XMin) / this._width;
var ky = (this.YMax - this.YMin) / this._height;

```

```

( )

```

```

: x1, y1, x1+w, y1+h
: x2, y1, x2+w, y1+h
: x1, y2, x1+w, y2+h
: x2, y2, x2+w, y2+h

```

```

var x1 = this.XMin + 6 * kx; //
var y1 = this.YMin + 4 * ky; //
var x2 = this.XMax - 26 * kx; //
var y2 = this.YMax - 25 * ky; //
var w = 20 * kx;
var h = 20 * ky;

```

```

6, 4, 26, 25, 20

```

ActionScript

```
var xms = this._xmouse;  
var yms = this._ymouse;  
if (xms >= x1 && xms <= x1+w)  
{  
    ... and so on  
}
```

ActionScript

4.2 *Scaleform*

Scaleform

Scaleform

scale9grid_window2.swf

scale9grid_window2.fla

ActionScript

, TitleBar

Border 1 4 (), Corner 1 4 ()

. Scaleform Scale9Grid

. ActionScript . OnResize()

```
import flash.geom.Rectangle;
//trace(this.scale9Grid);
this.MinW = 100;
this.MinH = 100;
this.XMode = 0;
this.YMode = 0;
```



```

this.OldX = this._x;
this.OldY = this._y;
this.OldW = this._width;
this.OldH = this._height;
this.OldMouseX = 0;
this.OldMouseY = 0;

function AssignResizeFunction(mc:MovieClip, xMode:Number, yMode:Number)
{
    mc.onPress      = function() { this._parent.StartResize(xMode, yMode);
    mc.onRelease    = function() { this._parent.StopResize(); }
    mc.onReleaseOutside = function() { this._parent.StopResize(); }
    mc.onMouseMove  = function() { this._parent.OnResize(); }
}

AssignResizeFunction(this.Border1, 0, -1);
AssignResizeFunction(this.Border2, 1, 0);
AssignResizeFunction(this.Border3, 0, 1);
AssignResizeFunction(this.Border4, -1, 0);
AssignResizeFunction(this.Corner1, -1, -1);
AssignResizeFunction(this.Corner2, 1, -1);
AssignResizeFunction(this.Corner3, 1, 1);
AssignResizeFunction(this.Corner4, -1, 1);
this.TitleBar.onPress      = function() { this._parent.startDrag(); }
this.TitleBar.onRelease    = function() { this._parent.stopDrag(); }
this.TitleBar.onReleaseOutside = function() { this._parent.stopDrag(); }

function StartResize(xMode:Number, yMode:Number)
{
    this.XMode = xMode;
    this.YMode = yMode;
    this.OldX = this._x;
    this.OldY = this._y;
    this.OldW = this._width;
    this.OldH = this._height;
    this.OldMouseX = _root._xmouse;
    this.OldMouseY = _root._ymouse;
}

function StopResize()
{
    this.XMode = 0;
    this.YMode = 0;
}

```

```

function OnResize()
{
    var dx = _root._xmouse - OldMouseX;
    var dy = _root._ymouse - OldMouseY;

    if (this.XMode == - 1)
    {
        this._x = this.OldX + dx;
        this._width = this.OldW - dx;
        if (this._width < this.MinW || _root._xmouse > this.OldX + this.OldW)
        {
            this._x = this.OldX + this.OldW - this.MinW;
            this._width = this.MinW;
        }
    }
    if (this.XMode == 1)
    {
        this._width = this.OldW + dx;
        if (this._width < this.MinW || _root._xmouse < this.OldX)
            this._width = this.MinW;
    }
    if (this.YMode == - 1)
    {
        this._y = this.OldY + dy;
        this._height = this.OldH - dy;
        if (this._height < this.MinH || _root._ymouse > this.OldY + this.OldH)
        {
            this._y = this.OldY + this.OldH - this.MinH;
            this._height = this.MinH;
        }
    }
    if (this.YMode == 1)
    {
        this._height = this.OldH + dy;
        if (this._height < this.MinH || _root._ymouse < this.OldY)
            this._height = this.MinH;
    }
}

```

draw primitive

draw

primitive

ActionScript

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tween

```

        Scale9Grid    SWF
    .   scale9grid_window2 fla   .   "TitleBar"
        Scale9Grid
    .
    .
Scale9Grid   .   "Save and Compact"   .
    .
        ActionScript   Scale9Grid   .
    .
    .

```

```

trace(this.scale9Grid);
    .
    .

```

```

this.scale9Grid = new Rectangle(24, 35, 363, 138);
    .
    .

```

```

        scale9grid_window3 fla   scale9grid_windoe3.swf   .
    .
    .

```

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        Scaleform
    .
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```

        Scale9Grid
Scaleform
scale9grid
    .
    .

```